

Mohammed Aljomaily

Software Engineer | AI & Cloud Technologies

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Professional Summary

Software Engineer specializing in cloud-native, AI-driven systems on Microsoft Azure. Built and operated production media pipelines covering transcription, translation, and automated document generation for enterprise clients, using event-driven Python services across Azure Container Apps, Cosmos DB, and Cognitive Services.

Experience

Software Engineering Intern – Emerging Technologies Department

Applab.qa, Doha, Qatar (client: Al Jazeera Media Network)

March 2025 – April 2026

- Primary author of the backend for Al Jazeera's AI video editing and multilingual translation platform, delivering the API service, the asynchronous job worker, and the projects API.
- Architected an event-driven pipeline across Azure Container Apps and Storage Queues, with a decoupled worker consuming queued work against Cosmos DB state.
- Built failure handling for a long-running distributed pipeline: poison-message routing, bounded retry with backoff, and per-action state logging.
- Integrated Azure Cognitive Services Speech and Azure OpenAI, supporting transcription and translation across 14 locales.
- Worked across Azure Functions, App Service, and API Management (APIM) alongside Semantic Kernel.

Key Projects

- **General Authority of Customs:** Built an AI chatbot from scratch on FastAPI and Semantic Kernel, with Azure AI Search vector search for semantic retrieval and a custom conversation history layer.
- **Invest Qatar:** Built the backend converting AI-generated structured JSON into formatted DOCX files in-process with Python file I/O, at roughly 30 ms per document, avoiding the external API dependency the team had assumed was required.

Projects

- **University RAG Chatbot:** Retrieval-augmented chatbot over university documentation using LangChain, OpenAI embeddings, and MongoDB Atlas Vector Search, with an incremental ingest pipeline that content-hashes each page to insert, update, or delete only what changed. Companion Selenium scraper extracted 184 documents from a 200-entry sitemap.
- **TIPOT, PDF Organising Tool:** Dockerised Flask application for PDF page reordering and merging; received and merged a pull request from an external contributor.
- **Aircraft Engine Trend Analysis:** Analysed multi-model turbine engine trend data (747/767/777/787) in Python, correlating exhaust gas temperature, fuel flow, and N1/N2 vibration signatures across the fleet.

Technical Skills

Languages: Python, Go (Golang), C (Linux), TypeScript

Cloud & Infrastructure: Docker, Linux (Debian/Arch), Azure Container Apps, Function Apps, App Service, API Management, Blob Storage, Queue Storage, Cosmos DB

AI & Frameworks: Azure OpenAI, Azure Cognitive Services (Speech & NLP), Semantic Kernel, LangChain, FastAPI, Quart, MongoDB Atlas Vector Search

Data: Cosmos DB, MongoDB, Postgres, pandas

Hardware: PCB design and fabrication, ESP32/ESP8266, KiCad, FreeCAD, 3D printing

Education & Certifications

B.Sc. Computer Engineering – Istanbul Aydin University, Istanbul, Turkey

2025 – Present

IGCSE & A-Levels: Physics, Mathematics, Computer Science

PCEP: Certified Entry-Level Python Programmer

2026

Programming with Google Go Specialization – University of California, Irvine (Coursera)